Analysis of Explainable Artificial Intelligence Methods in Security Information and Event Management Systems

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***Abstract*— The present article engages with the ongoing discourse surrounding the implementation of artificial intelligence and explainable artificial intelligence within the domain of information security systems. In the context of escalating cyber threats and mounting data volumes, conventional defense methodologies are proving to be inadequate. Consequently, the integration of artificial intelligence has emerged as a pivotal strategy to enhance the efficacy of threat detection and response processes. The article delves into the intricacies of explainability in artificial intelligence decision-making, exploring explainable artificial intelligence techniques such as local interpretable model-agnostic explanations, shapley additive explanations, gradient-weighted class activation mapping, and layer-wise relevance propagation. These techniques facilitate a more profound comprehension of the rationale behind artificial intelligence models' decisions, thereby enhancing the interpretability of these models. The discourse further encompasses the trade-offs between the accuracy and explainability of artificial intelligence models, along with their potential applications in security information management systems to enhance the efficiency and quality of incident analysis. The integration of explainable artificial intelligence into information security systems has the potential to enhance user confidence in artificial intelligence technologies and optimize human-machine interaction.**

Keywords*— AI, XAI SIEM, cybersecurity, LIME, cyber threats, information security, threat detection, incident response.

I. INTRODUCTION

The development of information technology and the increasing amount of data processed in various information security systems have led to the growing importance of the application of artificial intelligence (AI) to improve detection and response to threats. With the rapid growth of cyber threats and sophisticated attacks, traditional defense methods often prove insufficient. Therefore, the implementation of AI in information security systems is not just relevant but necessary to ensure the reliability and security of information resources.

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The integration of diverse models and algorithms capable of analyzing voluminous data in real time is a critical aspect of AI implementation [1, 2]. However, a salient issue emerges in this context: the necessity of defining the data to be fed into the system and the desired outcomes. The data must not only be processed by the system, but also be interpretable to information security practitioners. This necessitates that professionals possess a profound comprehension of both the data itself and the algorithms that process it.

A salient trend in AI is the concept of explainable AI (XAI), which aims to create transparent and comprehensible models. XAI is an approach that furnishes users with the capacity to discern how and why certain decisions were made [3]. In the context of information security, this is of particular importance, as a lack of explainability can lead to misunderstandings of the results of data analysis and, consequently, to ineffective actions to respond to threats.

Additionally, a lack of explainability may cause security professionals to distrust AI systems, which in turn may reduce their willingness to use such technologies. Users need to have confidence that decisions made based on data analysis are valid and reliable. Consequently, the integration of explainable AI into information security systems not only fosters increased trust in the technology but also facilitates more effective interaction between humans and machines.

II. AI AND XAI MODEL

A critical consideration pertains to the distinction between AI models and explained AI. An AI model constitutes a sophisticated program that is trained on specific data sets to execute various functions, including classification, prediction, and pattern recognition. However, despite their potency and efficiency, such models frequently operate as "black boxes," impeding the comprehension of their internal logic and the rationales behind their decisions.

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In contrast to conventional AI models, the field of XAI aims to develop transparent and interpretable systems, thereby enabling users to comprehend the rationale behind the outcomes obtained. This objective assumes particular significance in domains such as information security, where decisions made based on data analysis can bear critical consequences [4] To this end, a range of methods have been developed to achieve this objective, each with its own distinctive characteristics and approaches.The following is a review of the primary methods:

- LIME (Local Interpretable Model-agnostic Explanations) is a method that allows the creation and implementation of complex AI models using a variety of data and multiple attributes, while ensuring the interpretability of the results. LIME works by creating simple linear models that approximate the behavior of a complex model in the neighborhood of a particular prediction, thereby allowing users to gain an intuitive understanding of which factors had the greatest impact on the outcome [5, 6].
- SHAP (SHapley Additive exPlanations) is a method based on game theory that offers a powerful tool for evaluating the contribution of each parameter or feature of the model to the final value. SHAP allows users to see how each feature affects the prediction, providing a deeper understanding of the model's performance. This is particularly useful for assessing the importance of features and identifying potential anomalies in the data [7, 8].
- GradCAM (Gradient-weighted Class Activation Mapping) is a method primarily employed for the visualization and interpretation of the outcomes of deep neural networks, particularly in the context of computer vision tasks.GradCAM facilitates the identification of the regions of an image that exert the most influence on the model's prediction by generating heat maps that reveal the areas of the image to which the model directed its attention [9, 10].
- LRP (Layer-wise Relevance Propagation) is a method that employs a gradient-based approach to calculate the relevance of each layer of a neural network to the prediction. This method focuses on analyzing the influence of each layer of the neural network on the final prediction.The LRP method works by distributing the contribution of each layer in reverse order, which allows users to understand how information flows through the model and which layers have the greatest influence on the final result [11].

III. Use of AI in SIEM systems

Contemporary security information management (SIEM) systems encounter substantial limitations in their capacity to respond expeditiously to incidents, a deficiency that can have grave ramifications for organizational security. This predicament is further exacerbated in the context of a constantly evolving cyber threat landscape, where prompt response is of paramount importance. A notable challenge arises when disparate SIEMs operate at disparate intervals, such as five minutes, which can prove to be a critical factor in incidents lasting a mere one minute.

For instance, an outage in network connectivity, occurring at any point in the infrastructure, could be classified as a threat incident. However, it proves challenging for analysts and the SIEM system itself to respond expeditiously and ascertain the nature of the threat, particularly when the threat is characterized by a single change in parameter or value. In this regard, the integration of artificial intelligence technologies within SIEM emerges as a promising domain for industry advancement.

Specifically, AI enables the automation of data analysis, the detection of anomalies and suspicious activity, and the determination of event essence and prediction of potential attacks. To address this challenge, a supplementary add-on for existing SIEM systems employing the most effective AI models can be developed. The purpose of this add-on would be to perform post-analysis of data collected by the SIEM system, thereby enabling information security professionals to more accurately ascertain the nature of the incident.

This approach is expected to enhance the efficiency of incident response and improve the quality of analysis, ultimately strengthening the protection of information systems.

In order to develop and implement the most effective artificial intelligence model, it is necessary to consider several key components:

- The incident group must be considered.
- The input attributes of information security (IS) events must be considered.
- A variety of AI models must be considered.

However, a problem arises in the implementation process due to the trade-off between model accuracy and explainability (Fig. 1).

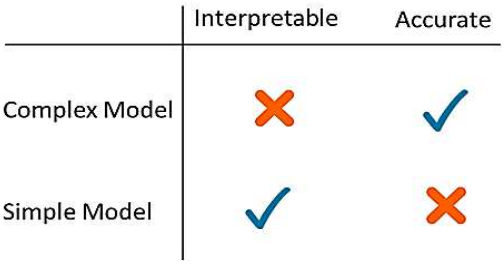


Fig. 1. Trade-off between the interpretability and accuracy of a complex and less complex model

The creation of an explainable incident card is accompanied by a potential loss of accuracy. This phenomenon arises from the proliferation of input data (incidents) and their multifaceted interactions with numerous variables (attributes of IS events) during the model training process. Consequently, the AI model evolves towards increased complexity and diminished predictability. It is evident that achieving a high degree of accuracy in results becomes challenging when nonlinear dependencies are employed (Fig. 2).

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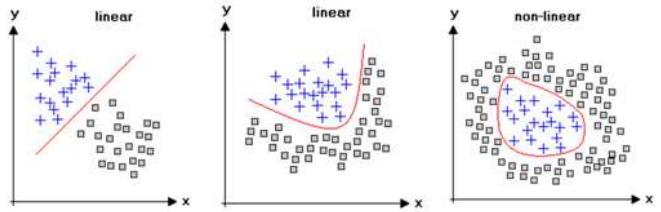


Fig. 2. Graphs of linear and nonlinear dependencies

In addressing the trade-off between accuracy and explainability of AI models, an XAI method called LIME can be effectively applied. All incidents entering the SIEM undergo a log collector, which, through the use of plug-ins, provides normalization of events. Initially, we used already normalized events to determine the most effective AI model. The LIME method addresses the problem of accuracy and explainability by considering incidents both before and after normalization. The training process of the model is carried out by partitioning the input data, representing the "complex model," into "linear models" (Fig. 3).

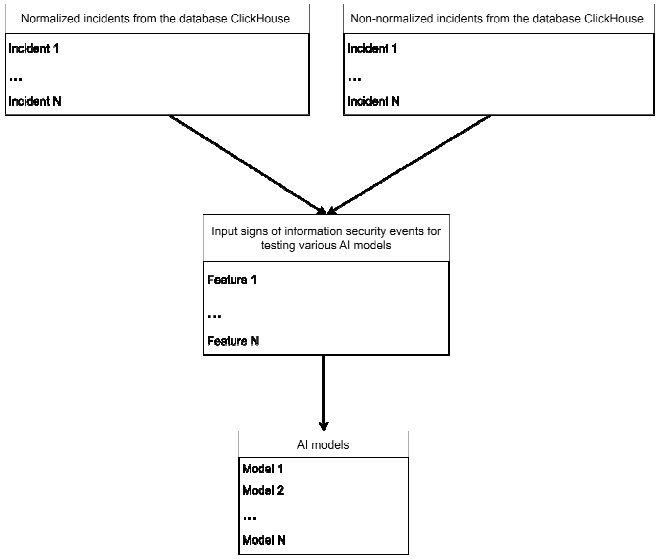


Fig. 3. Schematic of using the LIME XAI method with incident splitting

This partitioning method has been demonstrated to enhance the accuracy of the model while preserving its explainability. This enhancement is achieved by ensuring that each incident, whether normalized or non-normalized, traverses the same set of IS event attributes. The efficacy of this approach extends to the input attributes of IS events, enabling their partitioning into distinct components based on their characteristics. A notable illustration of this phenomenon is the analysis of normalized incidents involving the IS event "Linux. Incorrect password entry," where the relevant characteristics can be extracted (Fig. 4).

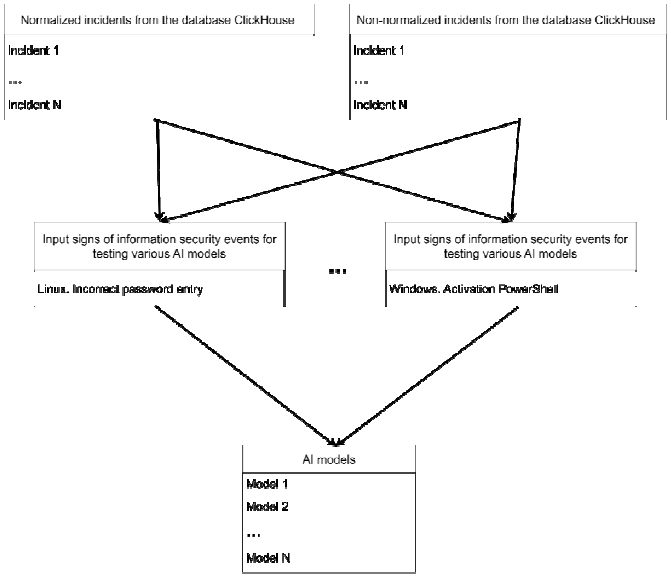


Fig. 4. Schematic of using the LIME XAI method with input features

IV. Implementing the XAI Add-on in a SIEM System

As part of the research, an experiment was conducted with a SIEM add-on based on an artificial intelligence model using LIME. The experiment used a distributed network with a virtual machine bank and a SIEM system. One of the subsystems lacked antivirus protection, resulting in a malicious virus being loaded onto one of the virtual machines. This incident was recorded and processed in both the standard SIEM and the SIEM with an integrated XAI-based add-on running in parallel (Fig. 5).

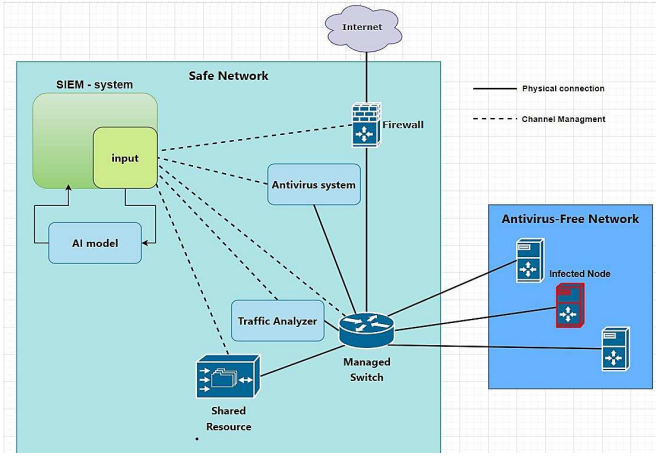


Fig. 5. Scheme of operation of SIEM with AI superstructure

The first scenario, using only the standard SIEM system, resulted in data showing an incident captured without the use of any additional tools (Fig. 6).

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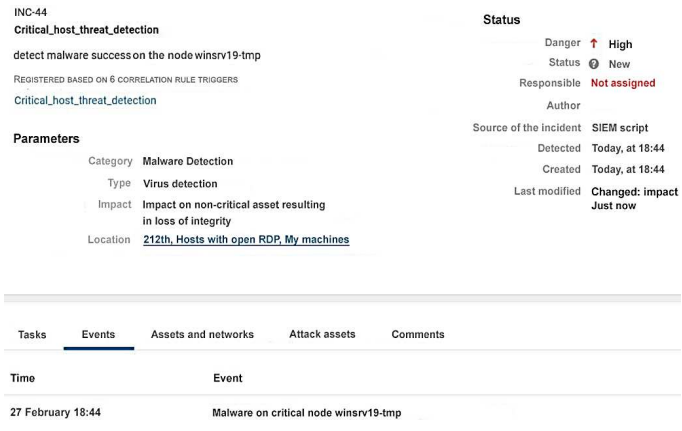


Fig. 6. Incident data to the SIEM system without the XAI-based add-on

In the second scenario, which involved a SIEM system with an XAI-based add-on, the data generated by the artificial intelligence was more meaningful (Fig. 7).

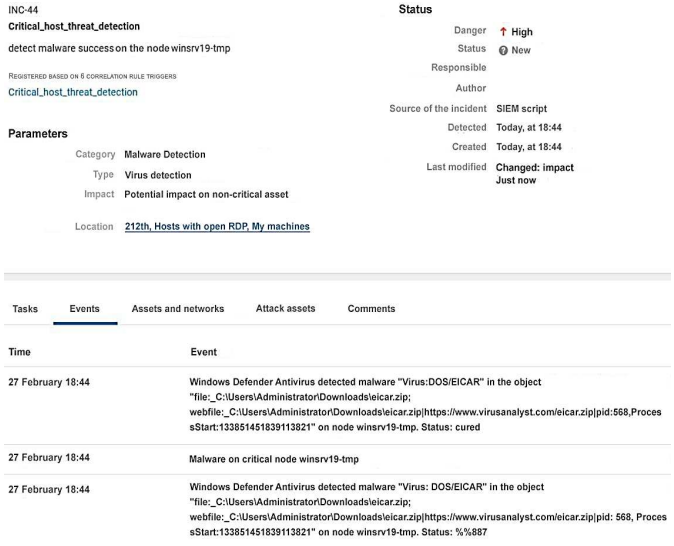


Fig. 7. Incident data in a SIEM with an integrated XAI add-on

Specifically, the system detected the presence of an antivirus program on a particular host and also recorded that the embedded security systems had successfully eliminated the threat, providing a complete description of the incident, including the virus name and location.

V. CONCLUSION

As the volume of data grows and the complexity of cyber threats increases, traditional defenses are becoming inadequate, making the adoption of artificial intelligence essential to improve threat detection and response. However, while highly accurate, many AI models operate as black boxes, making their solutions difficult to understand and reducing the confidence of information security professionals. In this context, XAI is becoming a key tool to improve the transparency and interpretability of AI models.

This article provides an overview of the major XAI methods such as LIME, SHAP, GradCAM, and LRP. These methods allow practitioners to better understand how and why an AI model makes certain decisions, which is particularly important in the context of information security, where errors can have serious consequences.

One of the key challenges addressed in the paper is the trade-off between accuracy and explainability of AI models. Complex models, such as deep neural networks, often provide high accuracy but are difficult to interpret. In contrast, simpler models may be more explainable, but their accuracy may be lower. To address this problem, the authors propose the LIME method, which allows complex models to be decomposed into simpler linear models while maintaining high accuracy and explainability.

The study conducted an experiment using a SIEM system augmented with an AI-based module using the LIME method. The experiment showed that integrating XAI into SIEM systems not only automates data analysis and anomaly detection, but also provides more detailed incident information. In particular, the system with the XAI module was able to not only record the incident, but also provide a complete threat description, including the virus name and location, which greatly simplifies incident analysis and response.

The experimental results show that the use of XAI in SIEM systems can significantly improve the efficiency of incident analysis and strengthen the protection of information systems. This is achieved by improving human-machine interaction and increasing expert confidence in AI-based decisions.

Thus, the integration of explainable AI into information security systems is a promising trend that can help organizations deal more effectively with today's cyber threats while ensuring the transparency and reliability of AI-based decisions.

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